Yanjin Li

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Education

University of Illinois Urbana-Champaign Sep 2024 – May 2027 (expected)
B.S. in Mathematics, Minor in CS Urbana, IL

Coursework: Deep Learning for Computer Vision, Applied Parallel Programming, Machine Perception, Numerical Analysis, Real Variables, Dynamical Systems, Probability and Statistics, etc.

Projects and Open Source

ScienceClaw | *Co-developer; Python* | github.com/beita6969/ScienceClaw ★ 1k stars ⑂ 110 forks (Oct 2026)

- Co-developed the replay-and-validation gate: reproduced repairs become a Skill patch plus typed Operators, promoted atomically as versioned snapshots only after boundary replay, hard-constraint, and noise checks.
- Integrated the ScienceClaw-Eval benchmark bridge into the core framework (IID/OOD evaluation, 23 disciplines).

Path Tracer and Graphics Engine | *C++, OpenGL, WebGL* | github.com/YJLi-new/RenderingPractice Sep 2024

- Wrote from scratch a Monte Carlo path tracer with BVH acceleration and a graphics engine in OpenGL and WebGL.

Publications

DecoupAttn: Decoupled Sparse Attention for Training-Free Acceleration of Video Diffusion
Yanjin Li, Bingbing Qiu, Qiang Huang, Rui Qian, Jian Wang, Mingda Zhang
In submission to CVPR 2027 arXiv:2611.05917

R^2 Flow: Recursive Self-Improvement via Recursive Skill Evolution
Mingda Zhang, Qiang Huang, **Yanjin Li**, Zijia Wang, Qika Lin, Xiaoying Tang, Tiesunlong Shen
Under review at ICLR 2027 arXiv:2609.33867

R^{2}Orch: Recursive Self-Improvement for Agent Orchestration via Self-Play within a Skill-Internalization Loop
Mingda Zhang, Yun Huang, Qiang Huang, **Yanjin Li**, Rui Qian, Qichao Zhao, Tiesunlong Shen, Na Zhao, Jian Wang, Xiaoying Tang
Under review at ACL Rolling Review arXiv:2610.22740

Research Experience

The Chinese University of Hong Kong, Shenzhen – T-Lab Apr 2026 – Present
Research Intern In collaboration with Mingda Zhang; group led by Prof. Xiaoying Tang

- **DecoupAttn:** ran the whole project **independently**, from idea and survey to experiments and writing.
  - Proposed training-free decoupled sparse attention for video diffusion transformers: semantic clusters compensate every key, and exact attention corrects their error on blocks fixed offline. Designed Distortion Budget and Reassign Cache, plus a fused compensation kernel on FlashInfer.
  - Benchmarked against **7 recent methods** on 10 video-generation, controlled-generation, and neural-rendering models: **SOTA** PSNR on 9 of 10 and up to **1.65**× faster than dense attention.
- **R^2 Flow:** wrote all of the core code and ran every experiment for the proposed method, including multi-GPU LoRA training of Qwen3.5-9B and a versioned skill library whose revisions pass independent checks.
  - Built an SGLang pipeline that overlaps agent rollout generation with gradient computation.
  - Built a **12-benchmark** evaluation suite that separates infrastructure failures from model errors.
  - Reached **SOTA** on all **12 benchmarks**, up to **+5.57 accuracy** points over the strongest baseline.
- **R^{2}Orch:** debugged the code of the proposed method, surveyed related literature, and ran the ablations.

Tsinghua University – Intelligent Connected Vehicles Research Group Sep 2025 – Dec 2025
Research Intern Advisor: Dr. Lei He

- Built an open-vocabulary 3D detection pipeline in MMDetection3D, fusing multi-view camera and LiDAR in BEV.
- Curated spatial instruction-tuning data from nuScenes and fine-tuned Qwen2.5-VL with LoRA to add open-vocabulary semantics to class-agnostic 3D proposals; ran evaluations and ablations on identical data splits.

Tsinghua University – Graphics and Geometry Computing Group Jun 2024 – Aug 2024
Research Intern Advisor: Prof. Kun Xu

- Benchmarked accuracy–efficiency trade-offs of light-transport algorithms for physically based and differentiable rendering (path replay backpropagation, importance sampling) on a C++/CUDA BVH-accelerated Monte Carlo path tracer.
- Built a reproducible C++/Python evaluation pipeline for convergence rate, memory footprint, and inverse-rendering quality; delivered an internal technical report and a baseline codebase.

Skills

Languages and tools: Python, C++, C, CUDA, OpenGL, WebGL, Git, Linux, LaTeX
ML and systems: PyTorch, PEFT/LoRA, SGLang, NCCL, FlashInfer, Weights & Biases, MMDetection3D